# The Quant Grimoire

Part I - Measure Theory (based on J.F. Le Gall)

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# Chapter 1

# Measure Spaces

This chapter lays the measure-theoretic foundation required to build the Itô integral. While standard texts like Le Gall focus heavily on pure mathematical rigor, our goal here is dual : to preserve that formal architecture while bridging the gap toward quantitative intuition. Before we can price a derivative or simulate a stochastic process in Python, we must first rigorously define the space of possible events and how we measure them.

> *"The starting idea of measure theory is to assign a non-negative real number (the measure of this subset) to each subset of a given set, in a way that satisfies certain natural additivity properties (the measure of a disjoint union must be the sum of the measures).*
>
> *For deep reasons, it is generally not possible to define the measure of any arbitrary subset, and one must restrict to a certain class of subsets, called **measurable sets**; a set equipped with a $\sigma$-algebra is called a **measurable space**."* — *J.-F. Le Gall [1]*

## 1.1 Measurable Sets and $\sigma$-Algebras

**Definition 1.1: $\sigma$-Algebra**

Let $E$ be a set. A collection $\mathcal{A}$ of subsets of $E$ is called a **$\sigma$-algebra** on $E$ if:

**(i) Certain Event :** $E \in \mathcal{A}$ ;

**(ii) Closed under Complementation :** $A \in \mathcal{A} \implies A^c \in \mathcal{A}$ ;

**(iii) Closed under Countable Unions :** If $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$, then $\bigcup_{n\in\mathbb{N}} A_n \in \mathcal{A}$.

Integration relies on the passage to the limit. We need to sum an infinite number of small pieces to perform limit calculations. The elements of $\mathcal{A}$ are called measurable sets, or $\mathcal{A}$-measurable sets if there is any ambiguity. We say that $(E, \mathcal{A})$ is a **measurable space**.

**Grimoire Insight: Intuition on $\sigma$-algebras**

The definition of a $\sigma$-algebra reflects three basic physical and logical constraints on what we can measure:

- **Measuring the whole space :** We must be able to measure the total size of our universe ($\Omega \in \mathcal{A}$).
- **Measuring the rest :** If we know how to measure an event $A \in \mathcal{F}$, we must automatically know how to measure "everything else". The measure of the complement $A^c$ is simply the measure of the total space minus that of $A$:

  $$\mu(A^c) = \mu(E) - \mu(A)$$

  Thus, $\mathcal{F}$ must be closed under complementation.
- **Passage to the limit :** Since analysis relies on continuous limits, if we can measure a sequence of individual events $(A_n)_{n\in\mathbb{N}}$, we must also be able to measure their combined occurrence ($\bigcup_{n=0}^{\infty} A_n \in \mathcal{A}$).

**Property 1.1.1: Immediate Properties of $\sigma$-Algebras**

Let $\mathcal{A}$ be a $\sigma$-algebra on a set $E$. The definition implies the following properties:

1. **Empty Set :** The empty set belongs to the collection :

   $$\emptyset \in \mathcal{A}$$

2. **Closure under Countable Intersections :** If $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$, then their intersection is also measurable :

   $$\bigcap_{n\in\mathbb{N}} A_n \in \mathcal{A}$$

3. **Closure under Finite Operations :** By choosing $A_n = \emptyset$ (or $A_n = E$) for all $n$ beyond a certain index, condition (iii) implies that $\mathcal{F}$ is automatically closed under **finite** unions and **finite** intersections.

**Proof.** We prove each assertion sequentially using axioms (i), (ii), and (iii) of a $\sigma$-algebra :

1. By axiom (i), the full space belongs to the collection ($E \in \mathcal{A}$). Applying axiom (ii), its complement must also belong to $\mathcal{A}$, so $E^c \in \mathcal{A}$. Since $E^c = \emptyset$, we directly conclude that :

   $$\emptyset \in \mathcal{A}$$

2. Let $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{A}$ be a sequence of measurable sets.
   - By axiom (ii), their complements are also measurable : $A_n^c \in \mathcal{A}$ for all $n \in \mathbb{N}$.
   - By axiom (iii), the countable union of these complements is measurable : $\bigcup_{n\in\mathbb{N}} A_n^c \in \mathcal{A}$.
   - Applying axiom (ii) again, the complement of this union is in $\mathcal{A}$ :

     $$\left(\bigcup_{n\in\mathbb{N}} A_n^c\right)^c \in \mathcal{A}$$

By De Morgan's laws, we have $(\bigcup_{n\in\mathbb{N}} A_n^c)^c = \bigcap_{n\in\mathbb{N}} (A_n^c)^c = \bigcap_{n\in\mathbb{N}} A_n$. Thus :

$$\bigcap_{n\in\mathbb{N}} A_n \in \mathcal{A}$$

3. Let $p \in \mathbb{N}^*$ and let $A_1, \dots, A_p \in \mathcal{A}$. We construct an infinite sequence $(A_n)_{n\in\mathbb{N}}$ by padding with empty sets :

$$\forall n > p, \quad A_n = \emptyset$$

Since $\emptyset \in \mathcal{A}$, all elements of the sequence $(A_1, \dots, A_p, \emptyset, \emptyset, \dots)$ belong to $\mathcal{A}$. By axiom (iii), the countable union is in $\mathcal{A}$:

$$\bigcup_{n\in\mathbb{N}} A_n = \left(\bigcup_{i=1}^{p} A_i\right) \cup \left(\bigcup_{n>p} \emptyset\right) = \left(\bigcup_{i=1}^{p} A_i\right) \cup \emptyset = \bigcup_{i=1}^{p} A_i$$

Therefore, $\bigcup_{i=1}^{p} A_i \in \mathcal{A}$. By taking complements and using De Morgan's laws again, finite intersections $\bigcap_{i=1}^{p} A_i \in \mathcal{A}$ follow immediately.

□

**Example 1.2: The Discrete/Maximal $\sigma$-Algebra**

Let $E$ be a non-empty set. The power set $\mathcal{P}(E)$, which contains all subsets of $E$, forms the maximal (or discrete) $\sigma$-algebra $\mathcal{A} = \mathcal{P}(E)$ on $E$.

**Verification.** To check that $\mathcal{P}(E)$ is indeed a $\sigma$-algebra :

(i) **Entire Space :** By definition, $E \in \mathcal{P}(E)$.

(ii) **Closure under Complements :** If $A \in \mathcal{P}(E)$, then $A \subseteq E$, so $A^c = E \setminus A \subseteq E$, which implies $A^c \in \mathcal{P}(E)$.

(iii) **Closure under Countable Unions :** If $(A_n)_{n\in\mathbb{N}} \subseteq \mathcal{P}(E)$, their union $\bigcup_{n\in\mathbb{N}} A_n$ remains a subset of $E$, hence $\bigcup_{n\in\mathbb{N}} A_n \in \mathcal{P}(E)$.

# Bibliography

[1] Jean-François Le Gall. *Intégration, Probabilités et Processus Aléatoires*. Springer, 2006.